

## ORIGINAL ARTICLE OPEN ACCESS

# Evaluating Higher Education Performance via Machine Learning During Disruptive Times: A Case of Applied Education in Türkiye

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## ABSTRACT

In response to the COVID-19 pandemic, an abrupt wave of digitisation and online migration swept the higher education institutions around the globe. In the aftermath of this digital transformation which endures as the legacy of the pandemic, what lacks in knowledge is how effective the anti-COVID measures were in maintaining quality education. Using machine learning to analyse student grades as a proxy for educational standards, this study investigates and demonstrates the evaluative potential of machine learning (vs. traditional statistics) with respect to not only crisis responses in education but also applied studies such as Information Systems and Tourism. Main implication of this study is the analytical utility of machine learning even when educational data are irregular and small. However, incorporating accurate and meaningful data points into the existing online educational systems is crucial to leverage this utility of machine learning.

## 1 | Introduction

UNESCO considers educational attainment as one of the main indicators of social sustainability (UNESCO 2017; UN 2022). In this sense, COVID-19 pandemic caused a major setback in sustainable development as the global average of higher education graduation rates, after two decades of steady increase, fell by 5% in 2020. Higher education institutions around the world shared the common concern for safeguarding the quality of their education from the disruptive effects of the pandemic (UNESCO 2024). Consequently, most universities around the globe hastily adjusted their instructive methods and transitioned to online learning systems. This transition mainly consisted of fully online classes on digital learning management platforms (Crawford et al. 2020; Montenegro-Rueda et al. 2021), videoconference tools (e.g., Zoom) or proctoring programs (e.g., Respondus).

Studies have been emerging that describe the instructional transformation and the evaluative changes universities have gone through during the pandemic (Crawford et al. 2020; Bao 2020; Toquero 2020; Yan 2020); however, the extent or the impact of these adjustments remains understudied (Karalis 2020).

Although the COVID-19 may have largely subsided, the legacy of the pandemic is substantial in terms of the digitisation of education systems permeating instructional platforms, assessment tools as well as administrative functions (Geisinger 2022; Markle 2023). Regardless of which types of learning platforms or assistive tools were chosen as anti-COVID measures, the rapid online transition as well as other restrictions imposed on education due to social distancing mandates indisputably created a challenging experience for most, if not all, university communities (Taylor et al. 2022). Hence, a salient gap in current studies is the effectual evaluation of anti-COVID

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measures universities took to combat pandemic's adverse effects on education. In this regard, this paper presents a unique study conducted to assess the effectiveness of anti-COVID measures using non-traditional statistical learning that is machine learning. By analysing data from two practice-oriented departments—Tourism and Hotel Management and Information Systems and Technologies—this study also makes a case for the utility of machine learning techniques in applied education. Understanding the effect of corrective measures might not only help to predict the effects of large-scale disruptions in education in the future (earthquakes, wildfires, floods, hurricanes, etc.) but may also be instrumental in appraising the overall effectiveness of post-covid digital transformations experienced in higher education.

## 2 | Background

### 2.1 | Pandemic-Time Higher Education

Tertiary education attainment rates have globally increased throughout the second millennial decade (UIS 2023), marking an average of 9% increase for OECD and an average of 6.7% increase for G-20 countries (OECD Data 2023) in 2020 when compared to 2010. However, the world witnessed a 5% decrease in overall higher graduation rates in 2020 (UIS 2023)—an undeniable impact of the COVID-19 pandemic. The pandemic also adversely affected international student enrolments in major higher education destinations such as the U.S. where international enrolments fell by 22% for the academic year of 2020–2021 as compared to the previous year (Baer and Martel 2020; Martel 2021). Educational attainment is deemed a global indicator of social sustainability, for which such disruptive effect of the pandemic has posed a serious threat (UN-SDG 2022). On the learner side, the shift to online learning highlighted existing educational inequities, as students from underprivileged backgrounds often lacked access to necessary technology and reliable internet connections (Crawford and Cifuentes-Faura 2022) exacerbating the digital divide and unequal learning opportunities. On the institutional side, the online shift exposed financial and infrastructural deficiencies in higher education systems when it comes to pedagogical as well as technological readiness (Crawford et al. 2020; Treve 2021; Coman et al. 2020; Sari and Nayır 2020).

Since the onset of the pandemic in early 2020, higher education institutions started making drastic changes in their instruction, evaluation, and assessment methods. Reviewing 20 countries around the world, Crawford et al. (2020) reported that the initial reaction of most universities was to transfer class contents to online platforms in line with the country-wide practices of social isolation and distancing. As traditional assessment methods, such as in-person exams, were disrupted due to the inability to conduct them safely, universities turned to alternative assessment methods, including online exams, open-book assessments, essays, and projects. This shift required instructors to rethink their evaluation strategies and implement measures to prevent academic dishonesty.

Online learning platforms, such as Moodle, Blackboard, Canvas were most commonly used during the pandemic (Montenegro-Rueda et al. 2021). University instructors used the tools or

functions of these platforms or other freely available online resources to carry out online assessments, discussion posts, and forums. Other main methods of assessment included the use of video-calling software, such as ZOOM or Skype (Haqien and Rahman 2020; Lee 2021; Razavi 2020; Serhan 2020), which allowed for tutorials, oral tests or served as a proctoring tool during exam times.

Although the studies have been accumulating regarding the instructional and evaluative changes transpired around the globe following the pandemic (Crawford et al. 2020; Bao 2020; Toquero 2020; Yan 2020), knowledge is still limited in terms of the scope or details of these changes (Karalis 2020; Wise, Kuhfeld, and Cronin 2022; Maatuk et al. 2022; Tsang et al. 2021; Kanetaki et al. 2021). Even more noticeable is the lack of studies that evaluate the effectiveness of anti-COVID measures adopted by universities to safeguard their educational integrity. The pandemic was particularly testing for the instructors in terms of their online education capacities (Sari and Nayır 2020). Those who were readily experienced in distant learning were arguably better prepared for the educational shift that most universities experienced during the pandemic (Appana 2008; Caris, Ferguson, and Gordon 2002; Mittal et al. 2022). However, those who were accustomed to traditional instruction methods (in-class, face-to-face) likely experienced difficulties adjusting to the 'new normal' (Taylor et al. 2022). Hence, the experience or aptitude of instructors in digital learning technologies might be a significant factor for the effectiveness of anti-COVID measures in universities. Additionally, the type of course (course content, structure, etc.) might be an explanatory element. Disciplinary differences could naturally influence pandemic adjustments (Slade et al. 2022; Treve 2021). Furthermore, preliminary findings of Shankar, Arora, and Binz-Scharf (2023) suggest that type of degree and course structure might influence the effects of digital transformation. Existing dearth of studies regarding the potential effects of course types call for an investigation into intradisciplinary differences (such as the theoretical vs. applied courses) as well interdisciplinary differences in the effectiveness of anti-COVID measures.

As recent studies suggest that the digital legacy of COVID in higher education may be permanent both in learning and administrative domains (Lehr and Vaughan 2023; Tafazoli and Boroujeni 2022; Pan and Tao 2022; Breslin 2021; Maatuk et al. 2022), the implications of understanding the effectiveness of digital transformations in higher education may be far-reaching (Tsang et al. 2021; Shankar, Arora, and Binz-Scharf 2023; Treve 2021). However, the abrupt changes that took place during the pandemic in questionably controlled settings make it particularly hard for evaluation of their impacts. Making quick decisions with resources available at the moment and *hoping for the best* characterised much of pandemic time education-scapes (Treve 2021). Therefore, irregularities in data and methods abounded, which further complicates attempts to analyse how effective each method was. Traditional statistical tools heavily rely on certain assumptions such as regular, normalised data and a general symmetry between units of data. Thus, to gain useful insights from pandemic-era actions, one might need to consider non-conventional methods of analysis.

## 2.2 | Machine Learning in Higher Education

While studying pandemic-era educational data is crucial for a realistic projection of the future of digital learning (Treve 2021; Shankar, Arora, and Binz-Scharf 2023), existing analytical tools may prove too rigid in the face of inconsistent, fluctuating and often too small data sets. Therefore, this study employs machine learning in its analyses of outcome measures. The irregularities in educational data particularly experienced during the pandemic render most tools of conventional statistics ineffectual. For sound application, traditional statistical tools—ANOVA, *t*-test, etc. —require certain assumptions to be validated such as normal distribution of data, homogeneity of variances, randomisation of samples as well as a sufficiently large sample size (Hair, Babin, and Anderson 2019). The prerequisites for such statistical tools are further removed from the reality of available data at times of crises such as COVID-19, where data are unique, time-stamped, and non-substitutable.

As opposed to traditional statistical tools, machine learning does not have the stringent requirements to analyse seemingly disjointed, fragmented and highly heterogeneous data. Playing a transformative in the field of statistics, machine learning offers advanced techniques, tools, and methodologies that enhance the way data are analysed, patterns are discovered, and predictions are made. Benefits of machine learning as a statistical learning paradigm include handling of big data (Chen, Mao, and Liu 2014), explaining non-linear relationships and predictive modelling (Hastie et al. 2009), detecting statistical anomalies (Chandola, Banerjee, and Kumar 2009), cluster and time series analysis (Jain 2010; Talagala, Hyndman, and Athanasopoulos 2023), and, perhaps more uniquely, adaptive learning from changing data (Alpaydin 2010). Machine learning has minimal prerequisite assumptions for the distribution of the data when compared to traditional statistical tools (Bzdok, Krzywinski, and Altman 2018). Moreover, the fact that operational gap between machine learning and statistical modelling are fast diminishing (Merghadi et al. 2020; Shobha and Rangaswamy 2018) and it makes machine learning an ideal paradigm to test educational effectiveness. Therefore, machine learning was chosen as the evaluative tool to analyse the student performance in this study.

Within the past few years, there has been several studies that show the evaluative utility of machine learning tools in higher education specific to student performance, such as student retention (Cannistrà et al. 2022; Cardona et al. 2020; Ho, Cheong, and Weldon 2021), student success (Ciolacu et al. 2018; Gray and Perkins 2019; Tseng et al. 2023; Yakubu and Abubakar 2022; Obeng 2023), and program completion (Langan et al. 2018). These studies provide evidence for the function machine learning that can serve as an ideal statistical medium for analysing educational data that are large or with inherent irregularities.

## 2.3 | Machine Learning Models and Performance Evaluation

As a subfield of artificial intelligence, machine learning models are essentially software applications or programs that simulate

human intelligence to predict outputs for given inputs without explicit intervention by humans (Mahesh 2020; El Naqa 2015). Machine learning models achieve this by searching data for any recognisable patterns that might explain their distribution or specific relationships that exist between inputs and outputs. Depending on the task in hand, machine learning models might be (a) supervised, as in designed to learn which inputs predict which outputs based on previously fed examples, (b) unsupervised, as in designed to recognise inherent patterns in a dataset without any predetermined outputs or labels, or (c) reinforced, as in designed to interact with a dynamic environment through a feedback system to test the aggregate optimality of its predictions (Mahesh 2020; Jordan and Mitchell 2015).

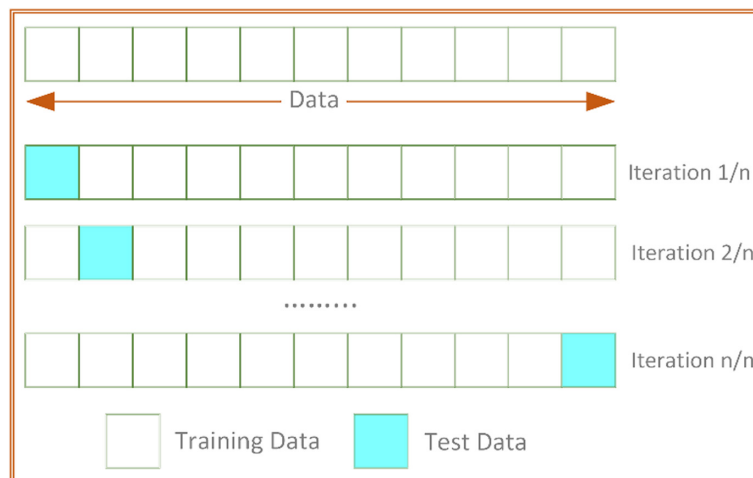
For the purposes of this study, supervised models that can accommodate small samples while safeguarding against overfitting were chosen. the characteristics of these models are explained subsequently.

### 2.3.1 | Random Forest (RF)

Regression and classification problems can both be solved with Random Forest—a technique for supervised machine learning that is based on decision trees (Žižka, Dařena, and Svoboda 2019). The forest is created by decision trees, and every tree has the same distribution. Throughout training, a random sample of data is utilised for feature selection and computation for every decision tree, which helps the algorithm to reduce noise and avoid overfitting.

### 2.3.2 | Leave-One-Out Cross Validation (LOO-CV)

While supervised learning models might excel in explaining any given data, they risk ‘overfitting’, that is, they lack prediction or generalisability over unseen, novel data (Berrar 2019). Cross-validation, hence, provides a safeguard against overfitting by splitting the data into two separate sets and tests the viability of any model inferred from one dataset by applying it on the other dataset (Wu and Hu 2024). Why did we choose unsupervised—cross validation—loo-cv <https://discovery.cs.illinois.edu/guides/Cross-Validation/Cross-Validation-in-Python/>. One common machine learning technique to measure the performance of any employed model is the “leave-one-out cross-validation” (LOO-CV) technique (Sammur and Webb 2011). In essence, leave-one-out cross-validation is a special case of cross-validation where the number of folds is equal to the number of instances in the dataset that is being used. The learning algorithm is applied once for each instance by using all other instances as the training set and the selected instance as a single-item test set. This procedure is depicted graphically in Figure 1. When the data size is small, LOO-CV is a better alternative since it employs more training samples every iteration, allowing the model to learn from the data more effectively. The best model can be found by employing grid search approach to discover the optimal hyperparameters. The open-source Python programming language (Van Rossum and Drake 2009a, 2009b) is a useful language which, with the help of the scikit-learn (sk-learn) library (Pedregosa et al. 2011), can perform all the required coding tasks.



**FIGURE 1** | Leave-one-out cross-validation visual representation.

### 2.3.3 | Data Preparation

Data scaling is a technique that is used during data preparation phase as it brings data points that are far apart closer to improve algorithm efficacy and speed up machine learning processing. It also helps to enhance algorithm performance and data quality by transforming the inputs into dimensionless or comparable distributions.

## 2.4 | Conceptual Framework and Contribution

Digitisation of the educational environment is the undeniable reality of 21<sup>st</sup> century and the expanse of digital learning tools necessitates instructors to be adequately proficient in online teaching (Carrier, Damerow, and Bailey 2017; Torrisi and Davis 2000). Instructional theories such as instrumental genesis stresses the importance of instructor's repeated interaction with technologies not only to improve technological knowledge but also to enhance overall instructional knowledge and preparedness (Clark-Wilson 2010; Clark-Wilson, Robutti, and Thomas 2020). Furthermore, the [connectivism](#) theory of learning implies that online savvy of instructors will reflect on students' learning competencies within their digital web of connections (Clark-Wilson, Robutti, and Thomas 2020; Duke, Harper, and Johnston 2013; Bates 2015; Goldie 2016). For the current study, these theoretical implications were central to the conceptualisation of the immediate effects of pandemic on educational performance as well as the aftermath observed in the following semesters. Namely, the authors recognised the fact that the sudden online transitioning and digitisation of courses during Spring 2020 left the instructors as well as the students with little to no time to prepare, and hence, forced them largely to rely on their existing online educational literacy. Any specific measures by universities to assist/manage the new digital format of education were arguably insufficient or less effective during this initial period compared to the following periods of pandemic (Fall semester of 2020 and so forth) where instructors and students had more time and resources to adapt to the *new normal*. Lastly, based on the recent studies (Slade et al. 2022; Shankar, Arora, and Binz-Scharf 2023), the content variation of the courses—specifically, the degree to which a course relied

on theory versus practical applications—was taken into consideration in this study. Selecting courses that varied in their theoretical/applied content allowed for a more diverse and realistic investigation of anti-COVID measures. Building on this framework, three main research questions guided this research study:

- How did anti-COVID measures impact student performance in the immediate vs. longer term?
- Did factors such as instructor's aptitude in online teaching and the course content play a role in this impact?
- Can machine learning techniques be viable tools to analyse educational data that may otherwise be unfit for conventional statistical tools?

## 3 | Methodology

### 3.1 | Study Setting and Data Collection

A private university in Türkiye was the study setting for this research study. This university was quicker than most universities in adopting its anti-COVID measures, such as fully transitioning to online classes the week following the lockdown announcement. The university also incorporated several techniques and tools into its adaptive instruction format—such as mirrors sent to each student's residence to be used to expose student screens during live exams, authentication photos taken by students in front of live class streams, and real time exam monitoring. However, as in other universities, these measures started in the middle of a semester which had started in regular format, resulting in significant adjustment and adaptation costs for the faculty, staff and students. Therefore, the measures taken during the first peak of COVID—which coincided with the ongoing Spring semester of 2020—were incomplete and limited in their effectiveness in countering the thoroughly unexpected impact of pandemic on higher education. This was largely deemed—by the university administration and faculty—the main reason for the unprecedented increase in overall grades for the Spring 2020 semester. Although student grades are by no means the only indicator of student performance—let alone quality of education, they were one of the few readily-available proxies in relation to



the educational impact of the pandemic in thoroughly chaotic times. In this sense, this study focused on student grades to evaluate, in broad terms, the scale of the impact of the pandemic as well as the anti-COVID measures taken by the university to counteract this impact.

As explained in the previous section, this study also aimed to address the gap in knowledge pertaining to how the effect of the pandemic and the subsequent measures might have varied across different types of course. Therefore, to ensure a diverse portfolio of courses were studied, courses were selected from two programs that had a high variation in their curricula in terms of course: Information Systems and Technologies (CTIS), and Tourism and Hotel Management (THM). The focal point in this selection was to capture different levels of ‘appliedness’ in courses which could in turn reflect differently on the overall performance of students after the digitisation of courses took place. Having distinct curricula from each other, selected programs had a commonality in that they housed a mixture of theoretical and applied courses at varying concentrations. The final set of courses chosen for this study comprised, from each program, one fully theoretical course (CTIS 163 *Discrete Mathematics* and THM 415 *Finance*), one partially applied course (CTIS 151 *Introduction to Programming* and THM 105 *Service Business Management*) and two fully applied course (CTIS 221 *Object Oriented Programming*, CTIS 487 *Mobile Application Development*, THM 246 *Restaurant Service*, and THM 323 *International Cuisines*). Prior to the study, an IRB approval (#2021\_09\_06\_03) was obtained from ‘Ethics Committee for Research Involving Human Subjects’ of Bilkent University, Türkiye, which also granted access to anonymous course grade data to be used for analysis. Please contact the authors for more details on the grade data analysed in this study.

### 3.2 | Machine Learning Models as Methodological Tools

The supervised machine learning technique used for this study was Random Forest (RF). Random Forest is a popular classification and regression algorithm with high accuracy (Hardman, Paucar-Caceres, and Fielding 2013; Rigatti 2017) and one that can perform better than other types of classifiers (Speiser et al. 2019; Parmar, Katariya, and Patel 2019). The student grades for various courses and semesters were assessed using the RF using ‘gini’ method of splitting the data. Prior to training and testing the model, all inputs were normalised by using the following equation:

$$X_n = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad 0 \leq X_n \leq 1 \quad (1)$$

where  $X_i$  is the measured data,  $X_n$  is the normalised data, whereas  $X_{\min}$  and  $X_{\max}$  represent the minimum and maximum data points, respectively.

The model was generated by training the data with ML algorithms to predict the output variables. Midterm, Final, Lab and Grades were used as inputs, and a binary variable representing a semester before or during COVID-19 was used as the output. To safeguard against overfitting, the unsupervised model of leave-one-out cross-validation (LOO-CV) method was also used to

**TABLE 1** | Class averages per course per semester.

| Course   | Spring 2019 | Spring 2020 | Spring 2021 |
|----------|-------------|-------------|-------------|
| CTIS     |             |             |             |
| CTIS 151 | 2.30        | 2.31        | 2.32        |
| CTIS 163 | 2.18        | 2.07        | 2.06        |
| CTIS 221 | 2.56        | 3.16        | 2.74        |
| CTIS 487 | 2.40        | 3.62        | 2.53        |
| THM      |             |             |             |
| THM 105  | 2.26        | 2.79        | 2.67        |
| THM 246  | 2.67        | 3.15        | 3.14        |
| THM 323  | 3.33        | 3.47        | 2.80        |
| THM 415  | 2.21        | 2.20        | 2.18        |

validate and measure the performance for the utilised model. Accuracy F1 Score, precision and recall were used to summarise the performance of the used classification algorithm. The optimum value for the hyperparameters was obtained by using grid search. The coding was performed in Python language. Sklearn library was used for the model training, fitting and testing purposes.

By using TP, TN, FP and FN measures, the confusion matrix map was utilised to evaluate the performance of the used model. Accuracy, recall, precision, and F1 scores were computed in accordance with Equations 2–5. The obtained results are tabulated in Table 1.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

### 3.3 | Analysis and Findings

Four courses from both Information Systems and Technologies (CTIS-coded) and Tourism and Hotel Management (THM-coded) departments during the Spring semesters of three consecutive years—Spring 2019–2021 – were selected for analysis. These semesters were chosen so that (a) comparison of Spring 2019 and Spring 2020 would reveal the change in grades once the pandemic hit the calendar and (b) comparison of Spring 2019 and Spring 2021 would reveal whether grades showed a return to normalcy. Respectively, these comparisons would serve as proxies for (a) the scope of the adverse of the pandemic on student performance and (b) the level of effectiveness of

anti-COVID measures fully in place. For reference, class averages for each course across three different Spring semesters are shown in Table 1.

### 3.4 | Results of Machine Learning Models

Supervised and unsupervised machine learning models can be used to solve problems based on classification, regression, and clustering algorithms. In this article, student grades data were assessed by using RF classification machine learning model. The optimised model was generated by training the data with ML algorithm to predict the output. Different course assessments (midterm, final, quiz and letter grade) were used as inputs, whereas semester before COVID-19 (0) and during COVID-19 (1) was used as the output. The coding used for output classification was:

- a. 2019 Spring (0) and 2020 Spring (1)
- b. 2019 Spring (0) and 2021 Spring (1)

Based on these considerations, the final analysis was conducted to reveal the results shown in Tables 2 and 3.

Two main patterns emerged from the results for CTIS-coded courses. First pattern was observed in two courses (CTIS 221 and 487) where outputs were significantly higher in Spring 2020 compared to those in Spring 2019 outputs, mirroring the widespread grade change due to the pandemic. For the following

semester of Spring 2021, the outputs of both these courses receded back to reveal no significant differences from Spring 2019. The second pattern reflected in the remaining two courses (CTIS 151 and 163) was that no significant differences occurred in outputs for any of the semesters.

The results for the THM-coded courses revealed more variation in how the grade outputs were classified. While one course (THM 415) showed no significant differences in outputs across the semesters, two courses (THM 105 and 246) revealed a pattern where increases in Spring 2020 did not recede in Spring 2021 and largely stayed the same. The remaining one course (THM 323) revealed the sole pattern of decreasing grades for Spring 2021 (compared to previous semesters), which was attributed to reduced confidence in the tests due extremely small input sizes (2–5 students per course per semester).

### 3.5 | Post Hoc Interviews With THM Faculty

To understand the pattern observed in THM courses better, a set of short interviews were conducted with the THM faculty. Six faculty members who instructed THM courses 105, 246, 323 and 415 across three semesters were interviewed. The interviews sought answers to three central questions: (a) How comfortable are you with digital instruction technologies? (b) How did you cope with online teaching (when the pandemic first started vs. after anti-COVID measures were taken)? and

**TABLE 2** | Confusion matrix and other performance scores for THM.

| Confusion matrix |        | Y-predicted |    | Performance scores |          |           |        |
|------------------|--------|-------------|----|--------------------|----------|-----------|--------|
| Model            | Y-true | 0           | 1  | Accuracy           | F1 score | Precision | Recall |
| THM              |        |             |    |                    |          |           |        |
| THM 105 (a)      | 0      | 46          | 11 | 0.819              | 0.824    | 0.817     | 0.831  |
|                  | 1      | 10          | 49 |                    |          |           |        |
| THM 105 (b)      | 0      | 39          | 18 | 0.688              | 0.685    | 0.679     | 0.691  |
|                  | 1      | 10          | 49 |                    |          |           |        |
| THM 246 (a)      | 0      | 7           | 2  | 0.769              | 0.667    | 0.600     | 0.750  |
|                  | 1      | 1           | 3  |                    |          |           |        |
| THM 246 (b)      | 0      | 8           | 1  | 0.786              | 0.667    | 0.750     | 0.600  |
|                  | 1      | 2           | 3  |                    |          |           |        |
| THM 323 (a)      | 0      | 11          | 1  | 0.895              | 0.857    | 0.857     | 0.857  |
|                  | 1      | 1           | 6  |                    |          |           |        |
| THM 323 (b)      | 0      | 11          | 1  | 0.895              | 0.857    | 0.857     | 0.857  |
|                  | 1      | 1           | 6  |                    |          |           |        |
| THM 415 (a)      | 0      | 12          | 0  | 1.000              | 1.000    | 1.000     | 1.000  |
|                  | 1      | 0           | 7  |                    |          |           |        |
| THM 415 (b)      | 0      | 12          | 0  | 0.813              | 0.400    | 1.000     | 0.250  |
|                  | 1      | 3           | 1  |                    |          |           |        |

Note: Y-predicted 0 = TP; Y-predicted 1 = FP; Y-true 0 = TN; Y-true 1 = FN.

**TABLE 3** | Confusion matrix and other performance scores for CTIS.

| Confusion matrix |        | Y-predicted |    | Performance scores |          |           |        |
|------------------|--------|-------------|----|--------------------|----------|-----------|--------|
| Model            | Y-true | 0           | 1  | Accuracy           | F1 score | Precision | Recall |
| CTIS             |        |             |    |                    |          |           |        |
| CTIS 151 (a)     | 0      | 28          | 8  | 0.771              | 0.772    | 0.818     | 0.766  |
|                  | 1      | 11          | 36 |                    |          |           |        |
| CTIS 151 (b)     | 0      | 23          | 13 | 0.796              | 0.843    | 0.797     | 0.895  |
|                  | 1      | 6           | 51 |                    |          |           |        |
| CTIS 163 (a)     | 0      | 29          | 0  | 1.000              | 1.000    | 1.000     | 1.000  |
|                  | 1      | 0           | 30 |                    |          |           |        |
| CTIS 163 (b)     | 0      | 22          | 7  | 0.868              | 0.898    | 0.863     | 0.936  |
|                  | 1      | 3           | 44 |                    |          |           |        |
| CTIS 221 (a)     | 0      | 31          | 4  | 0.842              | 0.791    | 0.810     | 0.773  |
|                  | 1      | 5           | 17 |                    |          |           |        |
| CTIS 221 (b)     | 0      | 35          | 0  | 0.966              | 0.955    | 1.000     | 0.913  |
|                  | 1      | 2           | 21 |                    |          |           |        |
| CTIS 487 (a)     | 0      | 29          | 2  | 0.932              | 0.889    | 0.857     | 0.923  |
|                  | 1      | 1           | 12 |                    |          |           |        |
| CTIS 487 (b)     | 0      | 27          | 4  | 0.788              | 0.667    | 0.765     | 0.813  |
|                  | 1      | 3           | 13 |                    |          |           |        |

Note: Y-predicted 0 = TP; Y-predicted 1 = FP; Y-true 0 = TN; Y-true 1 = FN.

(c) What kind of adjustments, if any, did you make to your grading (when the pandemic first started vs. after anti-COVID measures were taken)?

Responses to interview questions, first, revealed that instructors in THM, similar to those in CTIS, had varying degrees of experience/expertise in digital instruction. However, a major difference from CTIS was the challenge of transforming certain applied content. Based on their content, THM courses could be broadly divided into two categories: courses that were largely theory-based with minimal to no practical content versus those that heavily relied on practical content. Instructor responses indicated that theory-based classes, such as THM 415 (Finance), were comprised of class materials that were relatively compatible with online instruction as they were readily digitised (e.g., EXCEL codes). Any practical content in such courses were also structured in quantifiable formats that were easy to transfer to online platforms. Furthermore, instructors of such courses were likely to be experienced in online teaching tools as some of these tools were already incorporated into the course content such as Moodle assignments and online quizzes. Hence, a course like THM 415 showed a similar pattern to the courses in CTIS where instructors were knowledgeable about online teaching and did not experience a significant difference in class performance across semesters.

Interviews revealed that THM instructors of courses with heavy practical content—such as THM 246 (Restaurant Service) – faced a major challenge. Such courses typically

required students to perform physically, and in the case of THM 246, in real-life settings. For this course, students were regularly observed and evaluated while serving in department's own practice restaurant that was open to public. Therefore, THM 246 necessarily incorporated a high level of uncertainty and *randomness* (due to student-customer interaction). According to the instructors, this organic and 'high-touch' characteristic of the course was considered 'near impossible' to transfer to or replicate in an online learning platform with existing technologies, even after anti-COVID measures were well established. What inevitably transpired instead was that the course was transformed into a largely theoretical, inorganic version which lost its main source of challenge; that is, it is uncertain and random elements. Hence, the instructors attributed the increase in grades due to this new online version which replaced the testing of real-life skills in highly variable settings with a content that is partially derivative of other courses. Instructor responses for THM 105 (Service Business Management) and THM 323 (International Cuisines), both of which were practically involved courses that required real-life performances, observations, interviews and/or surveys with the industry, echoed similar constraints that stripped the courses of their novel elements. In this sense, THM 105 manifested the same grade pattern with THM 246, which was attributed a similar loss of challenge in the course content. The lack of similar grade pattern in THM 323, however, was largely attributed to the very small size of the classes across the semesters (2–5 students per class). Although machine learning tools are known for accommodating sample

sizes smaller than what is required by traditional statistical tests, the student numbers for this course were considered too small even for machine learning models for a conclusive output. Finally, the interviews revealed that instructors of THM 105, 246 and 323 considered themselves knowledgeable in online teaching tools and methods, and hence, did not see their digital instruction experience a significant factor for the fluctuations in student grades.

## 4 | Discussion

Machine learning has remarkably transformed the way we approach and obtain insights from collected data. It is helping us to build sophisticated applications even in irregular, non-parametric settings which may prove challenging to solve or altogether unsolvable with traditional statistical methods. In this study, we used machine learning techniques to compare student grade data as proxy to assess the immediate (and unexpected) impact of COVID-19 on student performance as well as the effectiveness of anti-COVID educational measures. Machine learning proved to be a useful analysis medium for uneven, non-homogenous data spread across several semesters due to the limitations during COVID caused. Specifically, through machine learning, the researchers were able to assess the magnitude of changes in student grades before and after anti-COVID measures, as an indicator for the effectiveness of these very measures. Findings implied that these measures started to work for certain applied courses, whose instructors were relatively less online-education savvy, only after adequate time and resources were available for them to adapt.

When the pattern in CTIS grades was interpreted, the disparity in online education experience emerged as the main point of distinction. Some professors were readily apt and up-to-date in online instruction methods and tools, and any other associated technology (software or hardware). Classes with such instructors (CTIS 151 and CTIS 163) revealed no significant changes in grade averages before and after COVID measures. This was in contrast to classes with instructors who had less experience with online instruction and/or limited familiarity with associated technology. In such classes, we found significant changes between grades before and after COVID measures. Specifically, the immediate effect of transitioning online – that which occurred in the midst of the Spring 2020 semester – was manifest in markedly higher grades for that semester. However, the grades receded back to pre-COVID normal in the following semester of Spring 2021 when the measures were more fully planned and applied, and instructors had likely more time and resources to prepare. Overall, the patterns observed in grade changes for the CTIS classes were in line with the findings of previous research (Appana 2008; Caris, Ferguson, and Gordon 2002; Mittal et al. 2022), signalling the benefits of prior instructor experience in effective application of digital learning tools and methods.

Interviews were conducted for better interpretation of the patterns observed in THM course grades, which did not follow those of CTIS courses. Interviews strongly suggested there were other variables additional to instructor's online education that factored into observed changes in grades. Although

THM courses share broadly the similar applied characteristics with CTIS courses, the nature of 'applied' might have differed for THM courses. Altogether applied THM courses of tourism classes were deemed to have more variability or randomness compared to applied courses in CTIS. First, there might be practice-oriented courses in THM that arguably rely less on—or assisted with—computerised technologies (such as food preparation) than others (such as hotel reservation systems). Instructors of such courses with practical elements (i.e., Service Business Management and International Cuisines) emphasised in the interviews the loss of "hands-on" learning experience when certain integral parts of the courses (e.g., production of various food items) had to be viewed online instead of being experienced in-person (e.g., in a practice kitchen). Solely relying on videos and images showcasing practical contents, which lack the necessary physical interaction with students, might make the subject matter too abstract for consistent and engaging learning experience for the students. This possibility might reduce the standardisation or automation of operational elements of a course, which then might complicate an otherwise smoother transition to online format. Second, some of the practical THM courses might incorporate or be structured around elements high in randomness—such as those requiring interactions with customers (e.g., restaurant service classes). Such elements are usually not possible to incorporate in a fully online course format, which can render the class more theoretical, removing the 'challenge' that comes with randomness. As revealed in our interviews, such factors to interact with instructors' online aptitude in a way that might compromise the effectiveness online instruction further, or even render instructor's online aptitude ineffectual. When practical courses that incorporate regular encounters with customers and/or on-site performance evaluation may most of its 'challenging' qualities, they might be reduced to a mostly hypothetical content delivered via traditional textual or audio-visual tools. As the instructor of the Restaurant Service course mentioned in the interviews, coping with the 'stress of real-time serving' is what makes such a course unique in that it gives students an invaluable opportunity to put accumulated knowledge into test, get realistic and instant feedback on their work and develop their crisis-management skills in the face of inevitable mistakes. In a purely online setting, such a class might become less distinctive than other courses and even repetitive in its theoretical, methodological and/or evaluative content. Hence, this overlap with other courses might result in increased grades for the course even when delivered by instructors highly apt in online instruction, an even so-called effective anti-COVID measures are fully applied.

## 5 | Conclusion and Implications

The global education environment experienced unprecedented changes as a result of the COVID pandemic (Geisinger 2021). Although the pandemic has ceased to be a global threat for the time being, the world continues to witness catastrophes that disrupt educational systems on a large scale. The devastating earthquake that hit Türkiye and Syria in February 2023 forced a months-long online-only curriculum for the schools during Spring 2023 (Alfuqaha et al. 2023), while the monsoon floods of Pakistan heavily disrupted access to education for nearly 3.5



million school children (Perry, Barón, and Dahlin 2023). Since such disasters often come with no warning, most educational responses go through little to no testing before becoming local or regional mandates by the governments. It is, therefore, crucial to find solutions to evaluate the effectiveness of such educational interventions as quickly and efficiently as possible to minimise adverse effects on educational outcomes.

In this study, we used machine learning techniques to compare student grade data as proxy to assess the immediate (and unexpected) impact of COVID-19 on student performance as well as the effectiveness of anti-COVID measures. One important implication of this study is that machine learning algorithms might be useful tools of analysis in educational settings where various models can be created and modified to address specific evaluative needs in crisis times. Moreover, the ability to work with small data allows machine learning models to be responsive to data as they are generated in real time, rendering them effective tools to detect issues early on—such as deviations in student grade averages—and help preventive action. However, an important consideration due here is that machine learning models require accurate data for sound interpretation. While machine learning helps researcher analyse data in a complex situation which violates the assumptions of traditional statistical analyses, it might need to be supplemented with other qualitative and/or quantitative techniques for best possible interpretation. The authors' consultation with instructors via post hoc interviews was one such technique. Based on these consultations, authors concluded that the abrupt adoption of anti-COVID measures by universities during the pandemic has likely interacted with instructors' varying levels of pre-existing experience and aptitude in online/digital learning platforms/tools. Additionally, unique 'applied' characteristics of each course—such as those seen in Tourism and Hospitality majors—might prevent broad generalisations as they might involve certain *factors of randomness or variability* related to their practical requirements, skills and competencies. Therefore, it might be necessary to take extra precautions, adopt or develop additional technologies to accommodate the unique characteristics of each applied course to retain its educational value.

Machine learning and other artificial intelligence-based technologies can also be useful here with their capacities to mimic the randomness of an interactive and dynamic setting—such as food and beverage service—in a computer simulation. It should again be noted here that the utility of machine learning models hinges on accurate and valid data. What the pandemic has shown us is that many crucial data points exist in online education that pertain to instructors' and students' behaviours but are not adequately accounted for. Utilising student grades as a proxy for student performance in this study stems from this lack of relevant data. Recording more precise, time-stamped online behaviour data—such as online search for information and help, educational screen time, response time to online prompts, connectivity trends with classmates and instructors, frequency of technical assistance sought/provided—may all be valuable data points to supplement student grades and attendance for a more comprehensive appraisal of student performance when digital learning takes precedence.

## 6 | Limitations and Future Considerations

To focus on the variables such as course content (i.e., theoretical vs. applied) only a small number of courses from two degree programs with high curricular diversity were chosen. While this allowed for intradisciplinary comparison, greater interdisciplinary comparisons may be achieved with future studies that involve more degree programs. This study also used Random Forest and Leave-one-out Cross Validation to demonstrate the utility of machine learning in analysing irregular educational data. Although both these methods are highly popular, they are by no means the only techniques available nor are they without their limitations. Random Forest might resist easy interpretation due to its black-box nature and can be a resource-heavy application. Its overfitting risk, however, was addressed in this study by utilising leave-one-out cross-validation. As newer, more robust machine learning techniques emerge, studies must be replicated in the future to ensure reliability.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

### References

- Alfuqaha, O. A., U. M. Al-Masarwah, R. I. Farah, et al. 2023. "The Impact of Turkey and Syria Earthquakes on University Students: Posttraumatic Stress Disorder Symptoms, Meaning in Life, and Social Support." *Behavioral Science* 13, no. 7: 587.
- Alpaydin, E. 2010. *Introduction to Machine Learning*. 2nd ed. MIT Press.
- Appana, S. 2008. "A Review of Benefits and Limitations of Online Learning in the Context of the Student, the Instructor and the Tenured Faculty." *International Journal on E-Learning* 7, no. 1: 5–22.
- Baer, J., and M. Martel. 2020. *Fall 2020 International Student Enrollment Snapshot*. New York, NY: Institute of International Education. <https://www.iie.org/publications/fall-2020-international-student-enrollment-snapshot/>.
- Bao, W. 2020. "COVID-19 and Online Teaching in Higher Education: A Case Study of Peking University." *Human Behavior and Emerging Technologies* 2, no. 2: 113–115. <https://doi.org/10.1002/hbe2.191>.
- Bates, A. W. 2015. *Teaching in a Digital Age: Guidelines for Designing Teaching and Learning*. Vancouver, Canada: BCcampus.
- Berrar, D. 2019. "Cross-Validation." In *Encyclopedia of Bioinformatics and Computational Biology*, edited by S. Ranganathan, M. Gribskov, K. Nakai, and C. Schönbach, 542–545. Oxford, UK: Academic Press.
- Breslin, T. 2021. *Lessons From Lockdown: The Educational Legacy of COVID-19*. London, UK: Routledge.
- Bzdok, D., M. Krzywinski, and N. Altman. 2018. "Machine Learning: Supervised Methods." *Nature Methods* 15, no. 1: 5–6.
- Cannistrà, M., C. Masci, F. Ieva, T. Agasisti, and A. M. Paganoni. 2022. "Early-Predicting Dropout of University Students: An Application of Innovative Multilevel Machine Learning and Statistical Techniques." *Studies in Higher Education* 47, no. 9: 1935–1956. <https://doi.org/10.1080/03075079.2021.2018415>.

- Cardona, T. A., E. A. Cudney, J. Snyder, and R. W. Hoerl. 2020. "Predicting Student Degree Completion using Random Forest." Paper presented at the 2020 ASEE Virtual Annual Conference Content Access.
- Caris, M. I. E. K. E., D. Ferguson, and G. Gordon. 2002. "Teaching Over the Web Versus in the Classroom: Differences in the Instructor Experience." *International Journal of Instructional Media* 29, no. 1: 61–67.
- Carrier, M., R. M. Damerow, and K. M. Bailey. 2017. *Digital Language Learning and Teaching: Research, Theory, and Practice, Global Research on Teaching and Learning English*. New York: Routledge.
- Chandola, V., A. Banerjee, and V. Kumar. 2009. "Anomaly Detection: A Survey." *ACM Computing Surveys* 41, no. 3: 1–58.
- Chen, M., S. Mao, and Y. Liu. 2014. "Big Data: A survey." *Mobile Networks and Applications* 19: 171–209.
- Ciolacu, M., A. F. Tehrani, L. Binder, and P. M. Svasta. 2018. "Education 4.0-Artificial Intelligence Assisted Higher Education: Early Recognition System With Machine Learning to Support Students' Success." Paper Presented at the 2018 IEEE 24th International Symposium For Design and Technology in Electronic Packaging (SIITME).
- Clark-Wilson, A., O. Robutti, and M. Thomas. 2020. "Teaching With Digital Technology." *ZDM Mathematics Education* 52: 1223–1242.
- Clark-Wilson, A. 2010. "How Does a Multi-Representational Mathematical ICT Tool Mediate teachers' Mathematical and Pedagogical Knowledge Concerning Variance and Invariance?" [Doctoral Dissertation, University of London] London, UK: UCL Discovery. <https://discovery.ucl.ac.uk/id/eprint/10019941/>.
- Coman, C., L. G. Țiru, L. Meseșan-Schmitz, C. Stanciu, and M. C. Bularca. 2020. "Online Teaching and Learning in Higher Education During the Coronavirus Pandemic: Students' Perspective." *Sustainability* 12, no. 24: 10367.
- Crawford, J., K. Butler-Henderson, J. Rudolph, et al. 2020. "COVID-19: 20 Countries' Higher Education Intra-Period Digital Pedagogy Responses." *Journal of Applied Learning & Teaching* 3, no. 1: 1–20. <https://doi.org/10.37074/jalt.2020.3.1.7>.
- Crawford, J., and J. Cifuentes-Faura. 2022. "Sustainability in Higher Education During the COVID-19 Pandemic: A Systematic Review." *Sustainability* 14, no. 3: 1879.
- Duke, B., G. Harper, and M. Johnston. 2013. "Connectivism as a Digital Age Learning Theory." *International HETL Review* 2013: 4–13.
- El Naqa, I., and M. J. Murphy. 2015. "What Is Machine Learning?" In *Machine Learning in Radiation Oncology: Theory and Applications*, edited by I. El Naqa, R. Li, and M. J. Murphy, 3–11. Cham: Springer International Publishing.
- Geisinger, K. F. 2021. "The Consideration of Admissions Testing at Colleges and Universities: A Perspective." *Applied Measurement in Education* 34, no. 4: 237–239. <https://doi.org/10.1080/08957347.2021.1987907>.
- Geisinger, K. F., ed. 2022. *College Admissions and Admissions Testing in a Time of Transformational Change*. New York, NY: Taylor & Francis.
- Goldie, J. G. S. 2016. "Connectivism: A Knowledge Learning Theory for the Digital Age?" *Medical Teacher* 38, no. 10: 1064–1069.
- Gray, C. C., and D. Perkins. 2019. "Utilizing Early Engagement and Machine Learning to Predict Student Outcomes." *Computers & Education* 131: 22–32. <https://doi.org/10.1016/j.compedu.2018.12.006>.
- Hair, J. F., B. J. Babin, and R. E. Anderson. 2019. *Multivariate Data Analysis*. Boston, MA: Cengage.
- Haqien, D., and A. A. Rahman. 2020. "Pemanfaatan zoom meeting untuk proses pembelajaran pada masa pandemic covid-19." *Susunan Artikel Pendidikan* 5, no. 1: 51–56.
- Hardman, J., A. Paucar-Caceres, and A. Fielding. 2013. "Predicting Students' Progression in Higher Education by Using the Random Forest Algorithm." *Systems Research and Behavioral Science* 30, no. 2: 194–203. <https://doi.org/10.1002/sres.2130>.
- Hastie, T., R. Tibshirani, J. H. Friedman, and J. H. Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Vol. 2. New York, NY: Springer.
- Ho, I. M. K., K. Y. Cheong, and A. Weldon. 2021. "Predicting Student Satisfaction of Emergency Remote Learning in Higher Education During COVID-19 Using Machine Learning Techniques." *PLoS One* 16, no. 4: e0249423.
- Jain, A. K. 2010. "Data Clustering: 50 Years Beyond K-Means." *Pattern Recognition Letters* 31, no. 8: 651–666.
- Jordan, M. I., and T. M. Mitchell. 2015. "Machine Learning: Trends, Perspectives, and Prospects." *Science* 349, no. 6245: 255–260. <https://doi.org/10.1126/science.aaa8415>.
- Kanetaki, Z., C. Stergiou, G. Bekas, C. Troussas, and C. Sgouropoulou. 2021. "Analysis of Engineering Student Data in Online Higher Education During the COVID-19 Pandemic." *International Journal of Engineering Pedagogy* 11, no. 6: 27–49.
- Karalis, T. 2020. "Planning and Evaluation During Educational Disruption: Lessons Learned From Covid-19 Pandemic for Treatment of Emergencies in Education." *European Journal of Education Studies* 4: 218953477. <https://doi.org/10.46827/ejes.v0i0.3047>.
- Langan, A. M., W. E. Harris, N. Barrett, C. Hamshire, and C. Wibberley. 2018. "Benchmarking Factor Selection and Sensitivity: A Case Study With Nursing Courses." *Studies in Higher Education* 43, no. 9: 1586–1596. <https://doi.org/10.1080/03075079.2016.1266613>.
- Lee, G. 2021. *Zoom Edtech Chief: Bricks-and-Mortar Universities Will Be Next to Face Digital Disruption*. Verdict. <https://www.verdict.co.uk/zoom-on-the-16bn-edtech-boom-where-ai-and-hybrid-realities-will-change-education-for-all/>.
- Lehr, A., and S. Vaughan. 2023. "Humanistic Leadership and Its Enduring Legacy in a Post-Covid-19 World." In *Moving Higher Education Beyond Covid-19: Innovative and Technology-Enhanced Approaches to Teaching and Learning*, edited by A. Visvizi, M. D. Lytras, and H. J. Al-Lail, 195–209. Leeds, UK: Emerald Publishing Limited.
- Maatuk, A. M., E. K. Elberkawi, S. Aljawarneh, H. Rashaideh, and H. Alharbi. 2022. "The COVID-19 Pandemic and E-Learning: Challenges and Opportunities From the Perspective of Students and Instructors." *Journal of Computing in Higher Education* 34, no. 1: 21–38.
- Mahesh, B. 2020. "Machine Learning Algorithms-A Review." *International Journal of Science and Research (IJSR)* 9, no. 1: 381–386.
- Markle, R. 2023. "College Admissions and Testing in a Time of Transformational Change." *Applied Measurement in Education* 36, no. 2: 132–136. <https://doi.org/10.1080/08957347.2023.2201705>.
- Martel, M. 2021. *Fall 2021 International Student Enrollment Snapshot*. Institute of International Education. <https://www.iie.org/publications/fall-2021-international-student-enrollment-snapshot/>.
- Merghadi, A., A. P. Yunus, J. Dou, et al. 2020. "Machine Learning Methods for Landslide Susceptibility Studies: A Comparative Overview of Algorithm Performance." *Earth-Science Reviews* 207: 103225. <https://doi.org/10.1016/j.earscirev.2020.103225>.
- Mittal, A., A. Mantri, U. Tandon, and Y. K. Dwivedi. 2022. "A Unified Perspective on the Adoption of Online Teaching in Higher Education During the COVID-19 Pandemic." *Information Discovery and Delivery* 50, no. 2: 117–132.
- Montenegro-Rueda, M., A. L.-D. La Rosa, J. L. S. Sánchez-Serrano, and J. Fernández-Cerero. 2021. "Assessment in Higher Education During the COVID-19 Pandemic: A Systematic Review." *Sustainability* 13, no. 19: 10509. <https://doi.org/10.3390/su131910509>.
- Obeng, A. Y. 2023. "Consequential Effects of Using Competing Perspectives to Predict Learning Style in E-Learning Systems." *Cogent*

- Education 10, no. 1: 2218960. <https://doi.org/10.1080/2331186X.2023.2218960>.
- OECD Data. 2023. "Education Attainment—Population With Tertiary Education—OECD Data." <https://data.oecd.org/eduatt/population-with-tertiary-education.htm>.
- Pan, M., and J. Tao. 2022. "Towards a Sustainable Multidimensional Approach to English Proficiency Proof in the Post-Pandemic Era: Learning From the Legacy of COVID-19." *Sustainability* 14, no. 9: 5568.
- Parmar, A., R. Katariya, and V. Patel. 2019. "A Review on Random Forest: An Ensemble Classifier." Paper Presented at the International Conference on Intelligent Data Communication Technologies and Internet of Things (ICICI) 2018, Cham, 2019.
- Pedregosa, F., G. Varoquaux, A. Gramfort, et al. 2011. "Scikit-Learn: Machine Learning in Python." *Journal of Machine Learning Research* 12: 2825–2830.
- Perry, F., J. D. Barón, and L. Dahlin. 2023. "How Are the Children of Pakistan's 2022 Floods Faring?" World Bank Blogs. <https://blogs.worldbank.org/endpovertyinsouthasia/how-are-children-pakistans-2022-floods-faring#:~:text=During%20the%20monsoon%20of%202022,being%20left%20behind%20and%20forgotten>.
- Razavi, L. 2020. "Students Like the Flexibility: Why Online Universities Are Here to Stay." *The Guardian*, May 27, 2020. <https://www.theguardian.com/education/2020/may/27/students-like-the-flexibility-why-online-universities-are-here-to-stay>.
- Rigatti, S. J. 2017. "Random Forest." *Journal of Insurance Medicine* 47, no. 1: 31–39. <https://doi.org/10.17849/insm-47-01-31-39.1>.
- Sammut, C., and G. I. Webb. 2011. *Encyclopedia of Machine Learning*. Boston, MA: Springer Science & Business Media.
- Sari, T., and F. Nayır. 2020. "Challenges in Distance Education During the (Covid-19) Pandemic Period." *Qualitative Research in Education* 9, no. 3: 328–360.
- Serhan, D. 2020. "Transitioning From Face-to-Face to Remote Learning: Students' Attitudes and Perceptions of Using Zoom During COVID-19 Pandemic." *International Journal of Technology in Education and Science* 4, no. 4: 335–342.
- Shankar, K., P. Arora, and M. C. Binz-Scharf. 2023. "Evidence on Online Higher Education: The Promise of COVID-19 Pandemic Data." *Management and Labour Studies* 48, no. 2: 242–249.
- Shobha, G., and S. Rangaswamy. 2018. "Machine Learning." In *Handbook of Statistics*, edited by C. R. Rao and T. S. Rao, 197–228. Amsterdam, Netherlands: Elsevier Science & Technology.
- Slade, C., G. Lawrie, N. Taptamat, E. Browne, K. Sheppard, and K. E. Matthews. 2022. "Insights Into How Academics Reframed Their Assessment During a Pandemic: Disciplinary Variation and Assessment as Afterthought." *Assessment & Evaluation in Higher Education* 47, no. 4: 588–605. <https://doi.org/10.1080/02602938.2021.1933379>.
- Speiser, J. L., M. E. Miller, J. Tooze, and E. Ip. 2019. "A Comparison of Random Forest Variable Selection Methods for Classification Prediction Modeling." *Expert Systems With Applications* 134: 93–101. <https://doi.org/10.1016/j.eswa.2019.05.028>.
- Tafazoli, D., and S. A. Boroujeni. 2022. "Legacies of the COVID-19 Pandemic for Language Education: Focusing on Institutes Managers' Lived Experiences." *Journal for Multicultural Education* 16, no. 1: 30–42.
- Talagala, T. S., R. J. Hyndman, and G. Athanasopoulos. 2023. "Meta-Learning How to Forecast Time Series." *Journal of Forecasting* 42, no. 6: 1476–1501.
- Taylor, D., M. Bearman, S. Scarparo, and M. K. E. Thomas. 2022. "The Emotional Transition to Online Teaching: Grief, Loss, and Implications for Academic Development." *International Journal for Academic Development* 27, no. 2: 176–190. <https://doi.org/10.1080/1360144X.2022.2083141>.
- Toquero, C. M. 2020. "Challenges and Opportunities for Higher Education Amid the COVID-19 Pandemic: The Philippine Context." *Pedagogical Research* 5, no. 4: em0063.
- Torrise, G., and G. Davis. 2000. "Online Learning as a Catalyst for Reshaping Practice—The Experiences of Some Academics Developing Online Learning Materials." *International Journal for Academic Development* 5, no. 2: 166–176. <https://doi.org/10.1080/13601440050200770>.
- Treve, M. 2021. "What COVID-19 Has Introduced Into Education: Challenges Facing Higher Education Institutions (HEIs)." *Higher Education Pedagogies* 6, no. 1: 212–227.
- Tsang, J. T. Y., M. K. P. So, A. C. Y. Chong, B. S. Y. Lam, and A. M. Y. Chu. 2021. "Higher Education During the Pandemic: The Predictive Factors of Learning Effectiveness in COVID-19 Online Learning." *Educational Sciences* 11, no. 8: 446.
- Tseng, C.-H., H.-C. K. Lin, A. C.-W. Huang, and J.-R. Lin. 2023. "Personalized Programming Education: Using Machine Learning to Boost Learning Performance Based on Students' Personality Traits." *Cogent Education* 10, no. 2: 2245637. <https://doi.org/10.1080/2331186X.2023.2245637>.
- UIS. 2023. "UIS Statistics." <http://data.uis.unesco.org/>.
- UN-SDG. 2022. *The Sustainable Development Goals Report 2022*. NY, USA: UN Publications. <https://data.oecd.org/eduatt/population-with-tertiary-education.htm>.
- UNESCO. 2017. *Education for Sustainable Development Goals: Learning Objectives*. Paris, France: UNESCO. <https://doi.org/10.54675/cgb9153>.
- UN. 2022. *United Nations Sustainable Development Goals Report 2022*. NY, USA: UN Publications. <https://unstats.un.org/sdgs/report/2022/>.
- UNESCO. 2024. *Global Education Coalition*. <https://gloaleducationcoalition.unesco.org/>.
- Van Rossum, G., and F. L. Drake. 2009a. *Python 2.6 Reference Manual*. Scotts Valley, CA: CreateSpace.
- Van Rossum, G., and F. L. Drake. 2009b. *Introduction to Python 3: Python Documentation Manual Part 1*. Scotts Valley, CA: CreateSpace.
- Wise, S. L., M. R. Kuhfeld, and J. Cronin. 2022. "Assessment in the Time of COVID-19: Understanding Patterns of Student Disengagement During Remote Low-Stakes Testing." *Educational Assessment* 27, no. 2: 136–151. <https://doi.org/10.1080/10627197.2022.2087621>.
- Wu, Q., and H. Hu. 2024. "Cross Validation in Python." <https://discovery.cs.illinois.edu/guides/Cross-Validation/Cross-Validation-in-Python/>.
- Yakubu, N., and A. M. Abubakar. 2022. "Applying Machine Learning Approach to Predict students' Performance in Higher Educational Institutions." *Kybernetes* 51, no. 2: 916–934.
- Yan, Z. 2020. "Unprecedented Pandemic, Unprecedented Shift, and Unprecedented Opportunity." *Human Behavior and Emerging Technologies* 2, no. 2: 110–112. <https://doi.org/10.1002/hbe2.192>.
- Žižka, J., F. Dařena, and A. Svoboda. 2019. "Random Forest." In *Text Mining With Machine Learning Principles and Techniques*, edited by J. Žižka, F. Dařena, and A. Svoboda, 193–200. Boca Raton, FL, USA: CRC Press.